NEP Haryana Excellence Awards 2026: Revised Server Requirements and Sizing Calculations

Application	NEP Haryana Excellence Awards 2026 web portal
Date	22-10-2026
Replaces	The sizing issued on 19-10-2026 (two production VMs, about 550 GB SSD and 1 TB backup)
Planning load	At most **150 applications** (about 40 last year) and **250–300 concurrent users** (HSHEC, 21-10-2026)

At 150 applications and 300 concurrent users, the requirement drops to **one production VM, one staging VM and 250 GB of backup**. That is about 40% fewer vCPUs and less than half the storage of the earlier proposal.

1. Revised requirement

#	Server	vCPU	RAM	Storage
1	**Production**: web server, application, PostgreSQL 16 database and Redis cache on one VM	**4**	**8 GB**	**60 GB** SSD system disk + **100 GB** SSD separate, expandable evidence disk
2	**Staging / UAT**: all components on one VM	**2**	**4 GB**	**60 GB** SSD
3	**Backup storage**, separate from the production VM	–	–	**250 GB**, expandable
	Total	**6** (was 10)	**12 GB** (was 20–24)	**220 GB SSD + 250 GB backup** (was ~550 GB + 1 TB)

- All servers: Ubuntu Server 24.04 LTS, 64-bit.
- The production VM must be **resizable to 6–8 vCPU / 12–16 GB within a few hours** if the load test or the first week of live use shows CPU or memory above 70%.
- Bandwidth: **50 Mbps dedicated or 100 Mbps shared**.

2. Calculation inputs

#	Input	Value used	Basis
I1	Applications	**150** (last year: 40)	Given by HSHEC
I2	Peak concurrent users	**300**	Given by HSHEC (upper end of 250–300)
I3	Evidence files per application	**50**	Measured: a complete demo application in the code has 46–48 evidence files, rounded up
I4	Average evidence file size	**3 MB**	Assumption. Upload limits are PDF 5 MB; images, DOCX and XLSX 25 MB
I5	Re-upload headroom	**× 1.5**	Assumption. Replaced files are kept as older versions
I6	Time between actions for one user	**10 s** at peak (20 s normal)	Assumption, deliberately high
I7	Response time / processor time per request	**0.2 s / 0.05 s**	Typical for this software stack (Django, PostgreSQL); to be measured
I8	Memory per application worker process	**150 MB**	Typical for this software stack; to be measured
I9	Deadline peak	40% of applications on the last day, over 8 hours; busiest hour 3× the average	Assumption, typical deadline behaviour
I10	Backup retention	**35 days**	To be confirmed by the data centre

3. Calculations

3.1 Requests

```
Peak request rate   = 300 users ÷ one action every 10 s   = 30 requests/second
Normal request rate = 300 users ÷ one action every 20 s   = 15 requests/second

Uploads on the last day = 150 × 40% × 50 files            = 3,000 files
Average upload rate     = 3,000 ÷ (8 h × 3,600 s)         = 0.10 files/second
Busiest-hour upload rate = 0.10 × 3                       = 0.31 files/second
```

30 requests per second means everyone clicking at once. This is deliberately high.

3.2 CPU

```
Application   = 30 requests/s × 0.05 s                    = 1.50 cores
Database      ≈ 40% of application CPU                    = 0.60 cores
Upload checks = 0.31 files/s × 0.5 s (PDF check, hash)    = 0.16 cores
Web server, cache and operating system                    ≈ 0.25 cores
                                                          ------------
Peak demand                                               ≈ 2.5 cores
```

VM size	Peak use	Assessment
2 vCPU	125%	Overloaded; the portal would slow down at the deadline
4 vCPU	**63%**	**Inside the usual 70% planning limit.** Normal load is about 33%

3.3 Concurrent requests

```
API requests in progress = 30 requests/s × 0.2 s          = 6
Room for uploads at the same time                          = 5
                                                           ---
Requests in progress at peak                               ≈ 11
```

Suggested configuration: **5 worker processes × 3 threads = 15 at once**, about 35% spare. The web server receives the whole upload before handing it to the application, so a slow user connection does not hold a worker.

3.4 Memory

Component	Calculation	RAM
Operating system and monitoring		1.00 GB
Application workers	5 × 150 MB	0.75 GB
Files held in memory (worst case)	15 at once × 25 MB × 2	0.75 GB
PostgreSQL	512 MB buffers + connections + maintenance	0.75 GB
Redis cache and Nginx web server		0.15 GB
Total at peak		**≈ 3.4 GB**

VM size	Use at peak	Assessment
4 GB	85%	Too little room; a burst of large uploads could crash the service
8 GB	**43%**	**Recommended.** The rest is used as disk cache, which keeps the whole database in memory

3.5 Evidence storage

```
Evidence = applications × files × average size × re-upload headroom
         = 150 × 50 × 3 MB × 1.5                          ≈ 34 GB
```

Case	Calculation	Size
Last year's volume	40 × 50 × 3 MB × 1.5	9 GB
Planning case	150 × 50 × 3 MB × 1.5	**34 GB**
Larger files (5 MB average)	150 × 50 × 5 MB × 1.5	56 GB

Allocated: 100 GB on an expandable disk, with an alert at 70%. It still fits if the average file turns out to be about 9 MB.

3.6 Database

The demo database holds 8 complete applications in 2.5 MB. Even at 150 applications × 5 MB, doubled for indexes and logs, the database is **under 2 GB**. It fits in memory on the same VM, so **a separate database VM is not needed**.

3.7 System disk (60 GB)

Item	Size
Ubuntu, packages, application and built front end	15 GB
PostgreSQL data and transaction logs	2 GB
Local copy of the latest backup before it is shipped off the VM	5 GB
Logs, 30 days compressed	2 GB
Temporary space for uploads	5 GB
Free space for upgrades (at least 50%)	31 GB
Total	**60 GB**

3.8 Backup storage

Evidence is backed up **daily**, not weekly. Weekly copies would lose up to a week of colleges' uploads if the disk failed near the deadline. Daily full copies would need about 1.2 TB, so the evidence uses a **deduplicating backup tool** (restic, borg or storage snapshots), which stores each file once plus about 20% overhead.

```
Database: daily compressed copy ≤ 0.5 GB × 35 days, plus transaction logs   ≈ 20 GB
Evidence: daily deduplicated backups, 34 GB × 1.2                          ≈ 41 GB
                                                                           --------
Planning case                                                              ≈ 61 GB
If the evidence disk fills (100 GB × 1.2 + 20 GB)                          ≈ 140 GB
```

Allocated: 250 GB, which covers a full evidence disk with about 45% to spare.

3.9 Bandwidth

Traffic in the busiest hour	Calculation	Rate
Evidence uploads	0.31 files/s × 3 MB × 8 bits	7.4 Mbps
Evidence downloads by reviewers (assumed equal to uploads)		7.4 Mbps
API traffic	30 requests/s × 20 KB × 8 bits	4.8 Mbps
First page loads (2.7 MB, 300 users in one hour)	300 × 2.7 MB × 8 ÷ 3,600 s	1.8 Mbps
Total		**≈ 21 Mbps**

So **50 Mbps dedicated or 100 Mbps shared** is enough. At 50 Mbps a 5 MB PDF takes under 1 second and a 25 MB file about 4 seconds.

3.10 Why one production VM is enough

- **The database is small** (under 2 GB) and fits in memory on the same VM. A second VM adds a server to patch and a network hop, but no capacity that is needed.
- **Availability does not get worse.** The old design also had a single application server and a single database server, so either one failing stopped the service. Recovery in both designs depends on backups (§5).
- **Scaling is simple.** The VM can be resized without code changes, and the database can move to its own server later if needed.
- **Evidence stays on its own disk**, so it can grow and be backed up separately.

4. Network and firewall

Direction	From → To	Port	Purpose
Inbound	Internet → VM 1	443/TCP	HTTPS portal and API
Inbound	Internet → VM 1	80/TCP	Redirect to HTTPS only
Inbound	Admin VPN / bastion → all VMs	22/TCP	SSH, key-based only
Local only	VM 1	5432, 6379, 8000	PostgreSQL, Redis, application; never exposed
Outbound	VM 1 → mail relay	587/TCP (or 465)	Verification and password-reset emails
Outbound	**VM 1, VM 2 → Cloudinary API (`api.cloudinary.com`)**	**443/TCP**	**Nomination document uploads. Without this rule those uploads fail.** Proxy acceptable.
Outbound	All VMs → package mirrors	443/TCP	Security updates
Outbound	All VMs → NTP	123/UDP	Time synchronisation

- Firewall, WAF or load balancer must allow request bodies of **at least 30 MB** (a 25 MB file plus form data).
- One `haryana.gov.in` sub-domain with a trusted SSL certificate, an authenticated **SMTP relay**, and SPF/DKIM/DMARC records for the sender domain.

5. Backup and recovery

The production server is a single VM with no automatic failover. Recovery depends on the following, which must be in place **before the submission window opens**.

Item	Requirement
Database backup	Daily backup plus continuous transaction-log (WAL) archiving, for point-in-time recovery
Evidence backup	**Daily** incremental backup with a deduplicating tool
Retention	35 days, stored off the production VM
Maximum data loss (RPO)	**1 hour** for the database, **24 hours** for evidence files
Recovery time (RTO)	**4 hours** to rebuild the service on a new VM from backup
VM snapshot	Just before the submission window opens, and before every deployment
Restore drill	A full restore practised on staging before go-live

6. Load test before go-live

The processor time per request (0.05 s) and memory per worker (150 MB) are typical values, not measurements. They must be confirmed before submissions open.

1. **Staging, 150 users:** a trial run to find errors and tune the test. Staging has half the capacity of production, so it would fail a 300-user test whatever production can do.
2. **Production, 300 users:** the real test, run before go-live while the VM holds no real data. Test data is removed afterwards.

Pass target	Value
Test scenario	300 users, each acting every 10 s: log in, open the assessment, save, upload a 3–5 MB PDF and a 25 MB image, download evidence
Response time	95% of requests under 1 s; uploads under 5 s, including Cloudinary uploads
Error rate	Under 1%
CPU and memory	Under 70% on the production VM

7. Assumptions to confirm

- **Average file size (3 MB)** is an assumption. If HSHEC can give the total size of last year's evidence for the 40 applications, dividing it by 40 gives the real size per application and replaces the estimate of 50 × 3 MB × 1.5 = 225 MB.
- **Processor time and memory per request** are typical values; the load test in §6 confirms them.
- **The production VM must be resizable at short notice** in case real use turns out heavier.

8. Points for confirmation

1. **Cloudinary:** approval of outbound HTTPS to `api.cloudinary.com`, and whether nomination documents may be stored with a third-party provider whose servers may be outside India. If not, the portal must be changed to keep them on the evidence disk, which needs development time before go-live.
2. **Recovery:** agreement to the 1-hour / 24-hour data-loss limits and the 4-hour recovery time; availability of VM snapshots.
3. **Resizing:** whether the production VM can be resized within a few hours during the submission window.
4. **Evidence size:** last year's total evidence size (see §7).
5. Hosting facility, and whether Ubuntu 24.04 LTS is permitted.
6. SMTP relay and sender address; who manages the sender domain's DNS.
7. Whether a security audit (e.g. a CERT-In empanelled auditor) is required before go-live.

9. Requested sequence

1. **Staging VM first**, for installation, email, the restore drill and the 150-user trial.
2. **Production VM and backup storage** once staging is validated.
3. **300-user load test on production**, then go-live.